Performance Evaluation Report

AI/ML Internship Task – Week 01 | Task 1: Image Classification Model

1. Project Summary

A Convolutional Neural Network (CNN) was developed using TensorFlow/Keras to classify handwritten digit images (0-9) from the MNIST dataset. The pipeline covered dataset preprocessing, CNN architecture design, model training with early stopping and learning rate scheduling, and thorough performance evaluation on a held-out test set.

2. Dataset

MNIST dataset: 70,000 grayscale images of size 28x28 pixels across 10 classes (digits 0-9). 60,000 images used for training (split 90/10 into training/validation), and 10,000 images held out as the independent test set. Pixel values were normalized to the [0, 1] range.

3. Model Architecture

A CNN with two convolutional blocks (Conv2D -> BatchNormalization -> MaxPooling -> Dropout), followed by a fully connected dense layer with dropout regularization and a softmax output layer for 10-class classification. Total trainable parameters: ~422,000.

4. Training Configuration

Parameter	Value
Optimizer	Adam (lr=0.001)
Loss Function	Sparse Categorical Crossentropy
Batch Size	256
Epochs	8 (with EarlyStopping & ReduceLROnPlateau)

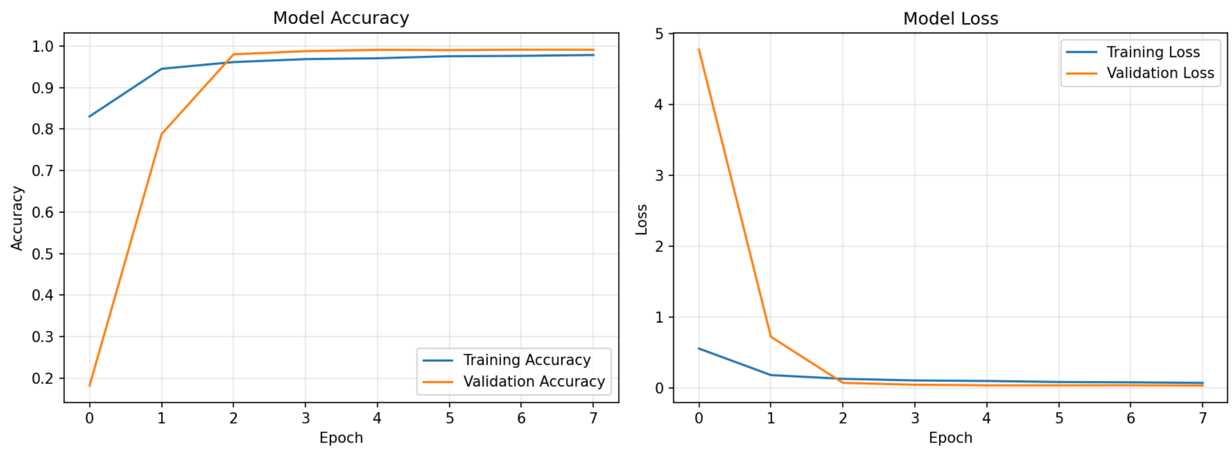
5. Evaluation Metrics (Test Set)

Metric	Score
Test Accuracy	99.02%
Precision (macro avg)	99.02%
Recall (macro avg)	99.00%
F1-Score (macro avg)	99.01%
Test Loss	0.0339

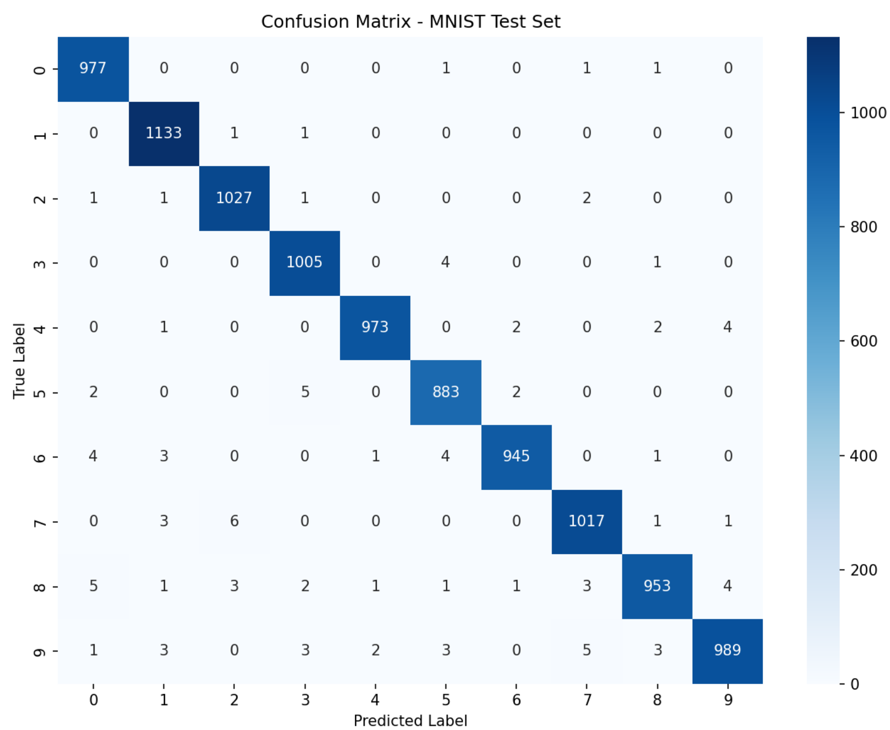
6. Per-Class Performance

Digit	Precision	Recall	F1-Score	Support
0	0.987	0.997	0.992	980
1	0.990	0.998	0.994	1135
2	0.990	0.995	0.993	1032
3	0.988	0.995	0.992	1010
4	0.996	0.991	0.993	982
5	0.985	0.990	0.988	892
6	0.995	0.986	0.991	958
7	0.989	0.989	0.989	1028
8	0.991	0.978	0.985	974
9	0.991	0.980	0.986	1009

7. Training & Validation Curves



8. Confusion Matrix



9. Results Analysis

The model achieved 99.02% accuracy on the held-out test set, with macro-averaged precision, recall, and F1-score all at approximately 99%. Training and validation accuracy curves converge closely with no significant divergence, indicating the model generalizes well and is not overfitting - aided by BatchNormalization and Dropout layers. The confusion matrix shows very few misclassifications, with the small remaining confusions occurring between visually similar digit pairs (e.g., 4 and 9, 3 and 5).

10. Suggested Improvements

- Introduce data augmentation (rotation, shift, zoom) to improve robustness to handwriting variation.
- Experiment with deeper or residual architectures for further accuracy gains.
- Apply advanced learning rate scheduling (e.g., cosine decay, warm-up).
- Ensemble multiple CNN models to boost overall accuracy.
- Test on out-of-distribution handwritten samples to evaluate real-world generalization.